

BAKALY, Russian Federation

WMO: 286150

Lat: 55.183N

Lon: 53.790E

Elev: 410

StdP: 14.48

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-18.1	-28.3	1.1	-23.5	-22.6	1.6	-18.0	28.0	11.8	25.8	14.7	2.8	350	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	87.3	67.6	83.9	66.7	80.6	65.0	70.1	83.1	68.6	80.6	66.8	77.4	7.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	96.6	74.8	64.2	91.3	73.5	62.7	86.5	71.8	34.2	82.9	33.0	81.2	31.6	77.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.0	19.0	-32.2	92.0	5.6	2.9	-36.2	94.1	-39.5	95.8	-42.6	97.4	-46.7	99.5
			-32.3	73.1	5.6	2.2	-36.3	74.6	-39.6	75.9	-42.7	77.1	-46.8	78.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	11.2	11.1	22.6	40.2	54.5	62.6	66.6	61.7	52.7	41.1	23.9	11.8	(6)
(7)		DBStd	23.02	13.66	15.13	9.16	10.58	8.94	7.54	6.34	7.54	8.35	8.67	11.41	14.58	(7)
(8)		HDD50	5865	1203	1090	849	330	57	3	0	2	64	301	783	1183	(8)
(9)		HDD65	9932	1668	1510	1314	745	340	135	59	158	380	742	1233	1648	(9)
(10)		CDD50	1657	0	0	0	34	197	380	515	364	144	23	0	0	(10)
(11)		CDD65	249	0	0	0	0	15	61	109	55	9	0	0	0	(11)
(12)		CDH74	2734	0	0	0	14	235	661	1152	566	106	0	0	0	(12)
(13)		CDH80	824	0	0	0	2	40	213	348	194	27	0	0	0	(13)
(14)	Wind	WSAvg	7.5	7.7	8.6	8.3	7.4	8.3	6.5	5.8	6.3	6.4	8.5	8.0	8.2	(14)
(15)	Precipitation	PrecAvg	18.3	1.1	0.8	0.8	0.8	1.4	2.2	1.7	2.1	1.5	1.9	1.3	1.2	(15)
(16)		PrecMax	24.5	2.1	1.6	1.5	1.9	2.9	3.7	5.0	6.3	3.4	3.9	3.0	2.2	(16)
(17)		PrecMin	10.6	0.0	0.0	0.1	0.0	0.3	0.6	0.2	0.4	0.2	0.2	0.0	0.1	(17)
(18)		PrecStd	3.9	0.5	0.4	0.5	0.6	0.8	1.0	1.3	1.3	0.9	1.1	0.9	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	36.9	44.0	76.0	84.4	91.4	90.8	84.7	68.7	43.5	35.6	(19)	
(20)		0.4%	MCWB	33.5	34.3	39.5	58.3	61.2	69.8	70.1	67.8	63.1	52.9	40.6	33.6	(20)
(21)		2%	DB	33.2	34.7	39.3	69.9	79.8	85.8	87.2	85.8	76.5	64.0	41.0	34.2	(21)
(22)		2%	MCWB	32.2	32.9	35.5	53.4	60.6	67.6	68.7	66.9	59.1	52.6	38.7	33.0	(22)
(23)		5%	DB	31.4	33.1	36.7	65.2	76.1	81.6	83.9	80.1	71.7	59.2	38.5	32.6	(23)
(24)		5%	MCWB	30.3	31.7	34.1	51.1	59.5	66.1	67.9	64.8	58.0	49.2	36.7	31.8	(24)
(25)		10%	DB	29.2	30.8	34.9	58.8	71.4	77.2	80.8	75.5	66.8	54.7	36.4	30.5	(25)
(26)		10%	MCWB	28.1	29.4	32.8	48.2	57.3	64.3	66.5	63.0	56.3	48.8	34.7	29.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	34.5	41.0	58.0	63.8	72.2	72.4	70.7	65.4	57.0	41.4	34.2	(27)
(28)		0.4%	MCDWB	34.5	35.8	44.5	75.2	77.7	86.1	84.2	83.0	78.5	63.6	42.5	35.0	(28)
(29)		2%	WB	32.4	33.1	36.1	54.4	62.0	69.5	70.6	68.2	61.9	54.4	39.1	33.1	(29)
(30)		2%	MCDWB	33.1	34.4	38.3	68.0	76.2	82.8	83.4	80.7	71.5	59.8	40.7	33.9	(30)
(31)		5%	WB	30.4	31.7	34.6	51.8	60.3	67.4	69.2	66.2	59.3	51.2	37.2	31.8	(31)
(32)		5%	MCDWB	31.3	33.1	36.4	63.6	74.3	78.7	81.4	77.3	68.7	58.2	38.2	32.6	(32)
(33)		10%	WB	28.0	29.4	33.0	48.1	58.4	65.2	67.8	64.0	57.1	48.5	35.1	29.6	(33)
(34)		10%	MCDWB	29.1	30.6	34.8	58.4	69.8	75.3	78.6	73.0	66.0	54.3	36.0	30.5	(34)
(35)	Mean Daily Temperature Range	MDBR	12.8	14.1	15.1	17.4	19.9	19.3	20.2	18.5	17.1	11.5	10.2	12.3	(35)	
(36)		5% DB	MCDBR	11.5	8.5	12.5	27.8	27.5	26.0	25.5	26.2	25.2	23.0	10.2	11.0	(36)
(37)		5% DB	MCWBR	11.6	7.9	10.0	13.7	12.1	10.9	9.6	9.8	11.9	13.2	8.1	10.9	(37)
(38)		5% WB	MCDBR	11.3	8.9	10.3	26.3	25.2	23.3	23.4	22.9	21.4	20.7	7.4	10.8	(38)
(39)		5% WB	MCWBR	11.4	8.4	8.7	14.1	11.8	10.5	9.6	9.5	12.4	12.8	7.4	10.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.319	0.371	0.413	0.399	0.394	0.411	0.433	0.381	0.354	0.317	0.292	(40)	
(41)		taud	2.042	2.030	1.960	2.092	2.272	2.336	2.290	2.206	2.323	2.283	2.183	2.021	(41)	
(42)		Ebn at Noon	187	230	244	254	267	270	262	246	245	220	181	155	(42)	
(43)		Edh at Noon	22	32	43	44	39	37	38	39	30	25	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	255	580	1038	1396	1719	1870	1841	1443	912	416	181	146	(44)	
(45)		RadStd	31	56	101	140	160	181	169	131	107	93	38	35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page